

DSM500 FINAL PROJECT · OCTOBER 2026

Synthetic Markets, Real Decisions

Evaluating the downstream utility of generative models for trading-strategy validation

Guillermo Enrique Olmos Ranalli

The problem

- A backtest tests a strategy on **one** historical path, so tuning many strategies finds lucky ones (backtest overfitting).
- Synthetic data promises **many** alternative histories.
- But generators are judged mainly on *realism* (sometimes on forecasting), rarely on the decisions they lead to.

Question and approach

Does a rule's rank on synthetic data predict its rank on **later, unseen** real data?



Spearman rank correlation (ρ) between synthetic and real 2022–2024 Sharpe ratios.

Study design

Data

USD/EUR daily
Build: 2018–2021
Test: 2022–2024

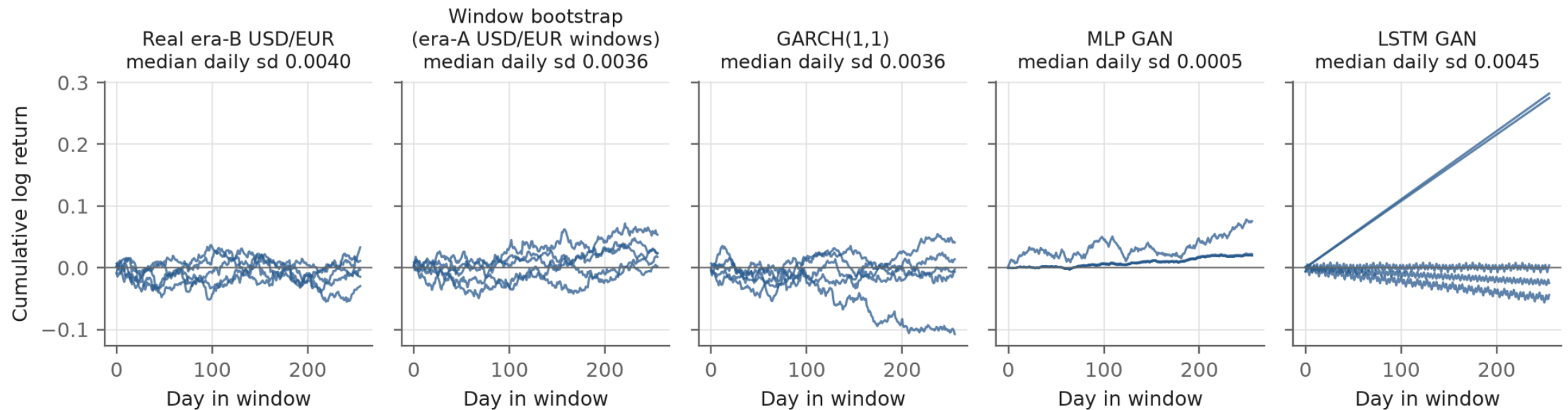
Generators

GARCH(1,1)
Window bootstrap
MLP GAN
LSTM GAN
1,000 paths each

Rules

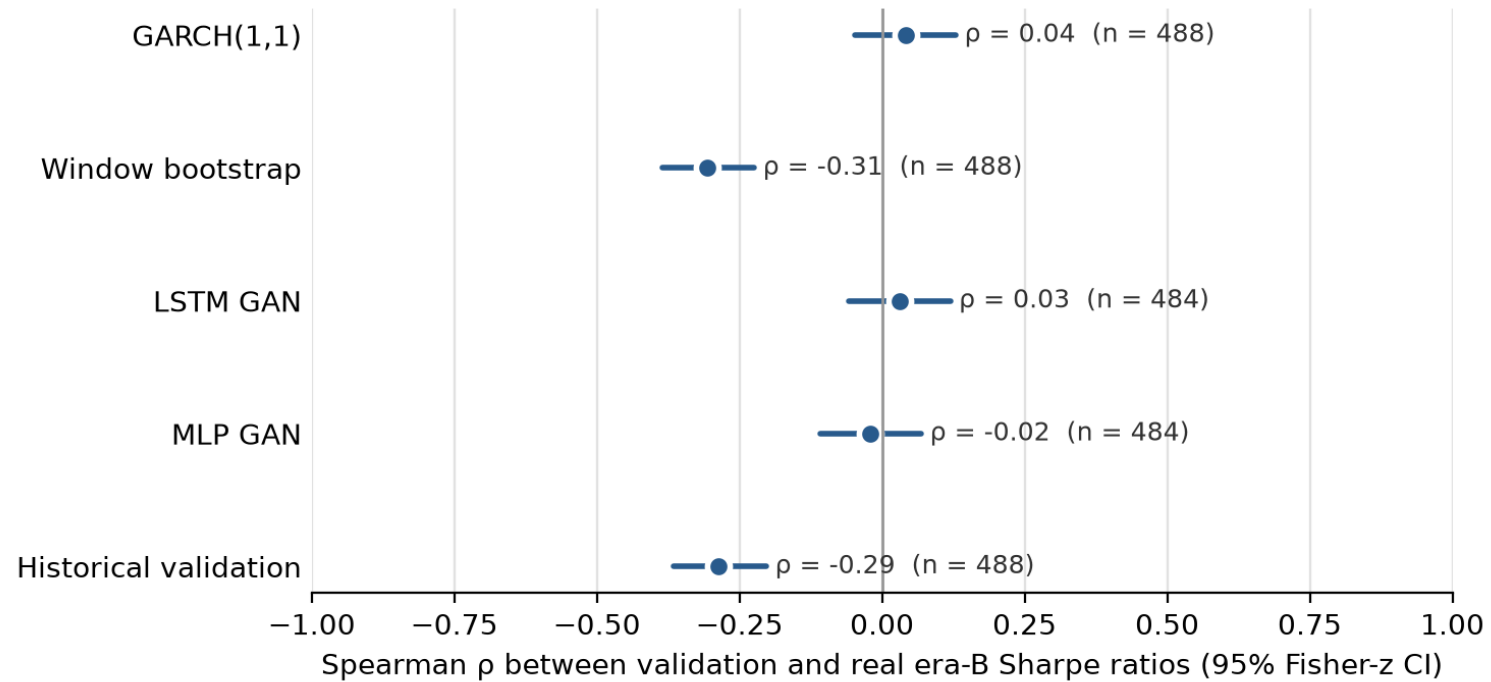
500 unique rules
480 moving-average
variants

What the paths show



Bootstrap matches history · GARCH calibrated · Saved GANs show fidelity limitations

Main result



GARCH $\rho \approx 0.04$ · LSTM ≈ 0.03 · MLP ≈ -0.02 · Bootstrap ≈ -0.31 · Historical ≈ -0.29

Scope and checks

- 484 rules shared by all five methods; 480 moving-average variants dominate the set.
- Paired resampling does not clearly separate GARCH from either GAN.
- Closely related rules may make the resampling intervals too narrow.
- Close agreement with training moments did not guarantee the best ranking.

Limitations

- Saved GANs retain pooled, zero-filled training; no new training in this rerun.
- Transaction costs and repeated training seeds are not evaluated.
- No synthetic ground-truth control; historical comparison uses one fixed split.
- One asset, one test period, mostly MA rules.

Corrected results regenerated: daily returns, unique rules and GARCH recursion fixed.

Conclusion

- No generator shows reliable positive ranking utility.
- Bootstrap and historical rankings are negatively associated with later performance.
- Close statistical fit did not guarantee the best ranking.
- Results apply to saved models and this USD/EUR experiment.

Next research: better trained models, broader rules and additional test periods.